EMBEDDING LATIN CUBES

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Abstract. The mathematical concepts of latin squares and latin cubes are introduced. We provide a perspective on when partial latin squares and partial latin cubes may be embedded within completed latin squares & cubes, respectively. These puzzles will be described from the perspective of known theorems about latin squares & cubes, as well as analogous graph theoretic frameworks.

1. Definitions and examples

The phrases "embedding in" and "extending to" are synonymous in this context, and will be used interchangeably throughout this work. When stating that an array Z has dimension $a \times b$, we mean that Z has a rows & b columns. When stating that an array Y of dimension $c \times d \times f$, we mean that Y has c rows, d columns, and f depths. Let the symbolism $j \in [k]$ be shorthand for $1 \leqslant j \leqslant k$ where $j, k \in \mathbb{Z}^+$. Furthermore, the degree of a vertex x in a graph G will be denoted by $deg_G(x)$. The maximum degree of a graph G is denoted by $\Delta(G)$.

If $t \in \mathbb{Z}^+$, a latin square of order t is a $t \times t$ array where the entries are the integers $i \in [t]$ and each of them occurs exactly once in each row and once in each column. More generally, the symbols can be any t distinct characters, so long as there is exactly one in each row and one in each column. Symbols from the english alphabet are used throughout, as well as the positive integers $\mathbb{Z}^+$, to populate latin squares and latin cubes. An example of a latin square is provided in figure 1.

1	2	3
2	3	1
3	1	2

Figure 1. A latin square of order 3

A partial latin square of order n is an $n \times n$ array where each symbol may occur, at most, once in each row and once in each column. Two examples are provided in figure 2. The latin square on the left cannot be completed. This is because the only missing entry in row two is 1, the only possible location for said 1 is in column one, yet there already appears a 1 in column one. The square on the right of figure 2 can be completed and the unique solution is the square shown in figure 1.

A latin rectangle of dimension $r \times s$ is an array with r rows and s columns where each entry is unequal to all other elements in that entry's row or column. An example of a 3×2 latin rectangle is shown in figure 3 below.

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1	2	
	3	2
	1	

1	2	
	3	1
	1	

FIGURE 2. Two partial latin squares of order 3

A	B
E	F
I	J

FIGURE 3. A latin rectangle of dimension 3×2

Consider an arbitrary latin square L of order n. Let us denote the symbol in L occupying the ith row and j^{th} column with l_{ij}. L is said to be "commutative" if and only if $l_{ij} = l_{ji}$ for all $i, j \in [n]$ where $i \neq j$. L is said to be "idempotent" if and only if $l_{ii} \neq l_{jj}$ for all $i \neq j$ where $i, j \in [n]$. The example on the left of figure 4 is both commutative and idempotent.

1	3	2
3	2	1
2	1	3

1	2	3	4
2	1	4	3
3	4	1	2
4	3	2	1

FIGURE 4. A commutative & idempotent latin square (left) and a purely commutative latin square (right)

Let $t \in \mathbb{Z}^+$. A latin cube of order t is a $t \times t \times t$ array filled with t^2 symbols such that each symbol appears exactly once in a given $t \times t$ "slice". A slice of the latin cube is obtained by holding one of the dimensions constant and collecting the entries that are in positions mutually orthogonal to this dimension. More formally, let us define an $n \times n \times n$ array and call it "C". We denote the symbol occupying the ith row, j^{th} column, and k^{th} depth of C with c_{ijk}. We say that C is a latin cube if and only if, for each integer $i \in [n]$, $c_{ijk} \neq c_{ilk} \neq c_{ijp}$ for all $j \neq l$, $k \neq p$ where $j, l, k, p \in [n]$. An example of a latin cube is provided in figure 5. Each vertically-oriented square is to be interpreted as a depth of the latin cube.

Let D be an array with r rows, s columns, and t depths (r,s,$t \in \mathbb{Z}^+$). Denote the symbol in D occupying the ith row, j^{th} column, and k^{th} depth with d_{ijk}. D is a latin rectangular prism if and only if both of the following conditions hold:

(1) $r \neq s$ OR $s \neq t$ OR $r \neq t$ OR $r \neq s \neq t$

(2) For each $i \in [r]$, $d_{ijk} \neq d_{ilk} \neq d_{ijm}$ where $j \neq l$, $k \neq m$ for j,$l \in [s]$ and k, $m \in [t]$.

An example of a $3 \times 4 \times 2$ latin rectangular prism is shown in figure 6.

depth 1

1	2	3
5	6	4
9	7	8

depth 2

4	5	6
8	9	7
3	1	2

depth 3

7	8	9
2	3	1
6	4	5

FIGURE 5. An example of a latin cube where each 3×3 grid is a depth

Let $n \in \mathbb{Z}^+$. A partial latin cube of order n is an $n \times n \times n$ array where each symbol may occur, at most, once in a given $n \times n$ slice. Two examples of partial latin cubes are provided in figure 7. The primary goal of this work is to discuss when a partial latin cube of arbitrary dimension can be extended to a completed latin cube.

A partial completion of the cube on the left of figure 7 is shown on the left of figure 8. The partial latin cube on the left of figure 8 cannot be completed further. One reason is because 1 & 7 are the only symbols missing in the first depth, yet both of these entries already appear in the second column of the cube. Thus, there is no way of filling in depth 1 without violating "latin-ness". A full completion of the cube on the right of figure 7 is shown on the right of figure 8. The authors are unaware of any published theorem demonstrating directly when an arbitrary latin cube can be extended to a completed latin cube. However, by conceptualizing a latin square as a bipartite graph or a latin cube as a tripartite hypergraph, it is possible to utilize results from graph theory to illuminate this issue. Defining and demonstrating this graph conceptualization is the purpose of the following two sections.

depth 1

A	B	D	E
C	F	G	I
H	K	J	L

depth 2

R	Z	L	J
B	H	K	D
G	Q	E	F

FIGURE 6. An example of a latin rectangular prism of dimension $3 \times 4 \times 2$

cube 1 depth 1

2		
		4
5		

cube 2 depth 1

1	2	3
5	6	4

cube 1 depth 2

	6	
	9	

cube 2 depth 2

4	5	6
8	9	7

cube 1 depth 3

1		3
	7	
		8

cube 2 depth 3

7	8	9
2	3	1

FIGURE 7. Two partial latin cubes of order 3

cube 1 depth 1

2	8	9
6	3	4
5		

cube 2 depth 1

1	2	3
5	6	4
9	7	8

cube 1 depth 2

4	6	7
8	1	5
	9	

cube 2 depth 2

4	5	6
8	9	7
3	1	2

cube 1 depth 3

1	5	3
9	7	2
		8

cube 2 depth 3

7	8	9
2	3	1
6	4	5

FIGURE 8. Partial completion of cube 1 in figure 7 (left) & full completion of cube 2 in figure 7 (right). Black symbols represent original entries, red symbols represent extended entries.

2. Latin squares as 1-factorizations of bipartite graphs

Let $p, q \in \mathbb{Z}^+$. A complete bipartite graph, with one part composed of p vertices and the other composed of q vertices, is denoted by $K_{p,q}$. A completed latin square of order n may be represented as a properly-colored $K_{n,n}$ if we let one part of $K_{n,n}$ represent row indices and the other part represent column indices. An entry in row i and column j of the latin square is represented by an edge in the bipartite graph connecting row vertex i to column vertex j, and the color of this edge corresponds to the symbol that fills this entry in the corresponding latin square. An example of a 3×3 latin square with its corresponding $K_{3,3}$ appears in figure 9 below. The 3×3 latin square in figure 9 corresponds to a nice coloring of $K_{3,3}$. In fact, this coloring of $K_{3,3}$ is a 1-factorization, since each color class induces a 1-factor. A 1-factor is a spanning subgraph where the degree of each vertex is exactly one.

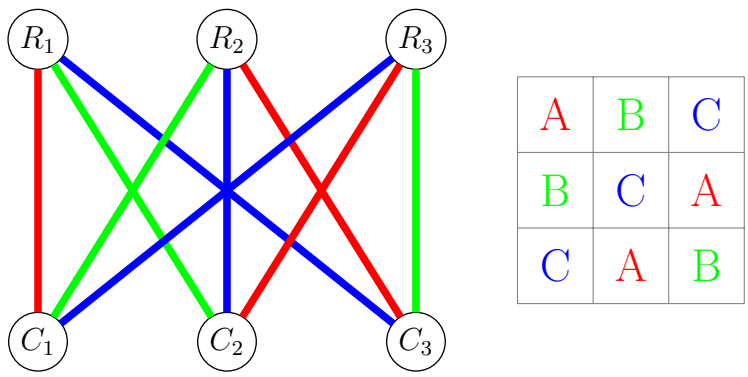

Figure 9. A 1-factorization of $K_{3,3}$ representing a commutative 3×3 latin square

Partial latin squares may be represented by a properly-colored bipartite graph that is not complete. Figures 10 & 11 show examples of 3×3 partial latin squares with their corresponding colorings of $K_{3,3}$. The latin square in figure 10 can be embedded in a completed latin cube, while the latin square in figure 11 cannot be embedded in a completed latin cube. Equivalently: the partial 1-factorization of $K_{3,3}$ in figure 10 can be extended to a 1-factorization of $K_{3,3}$ while the partial 1-factorization in figure 11 cannot be extended to a 1-factorization of $K_{3,3}$.

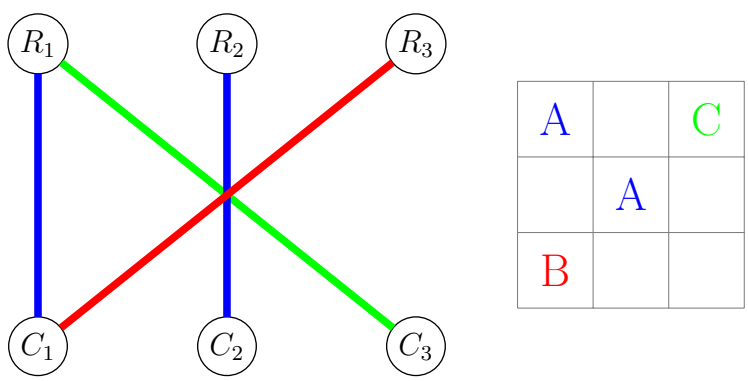

Figure 10. A 3×3 partial latin square with its corresponding partial 1-factorization of $K_{3,3}$. This partial latin square can be embedded in a complete latin square of order 3

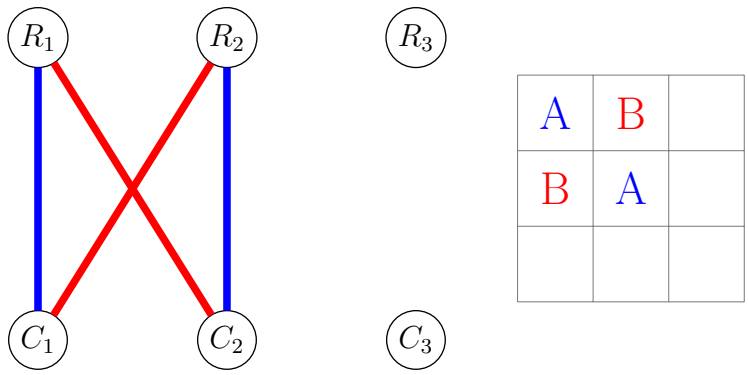

FIGURE 11. A 3×3 partial latin square with its corresponding partial 1-factorization of $K_{3,3}$. This partial latin square cannot be embedded in a complete latin square of order 3

3. Latin cubes as 1-factorizations of tripartite 3-uniform hypergraphs

Let $p, q, r \in \mathbb{Z}^+$. A complete tripartite 3-uniform hypergraph with its three parts composed of p, q, and r vertices is denoted by $K^3_{p,q,r}$. A 1-factorization of a hypergraph $\mathcal{F}$ is the same as a 1-factorization of a graph: it is a coloring of edges where each color class is a 1-regular subgraph and spans the vertex set of $\mathcal{F}$. A 1-factorization of $K^3_{n,n,n}$ corresponds to a latin cube of order n, since each part of the 3-uniform hypergraph may represent a dimension of the latin cube. More precisely, let $\mathcal{G} := K^3_{n,n,n}$. In the case that $\mathcal{G}$ is to represent a latin cube, the following labeling of its vertex set is useful: $V(\mathcal{G}) = \{R, C, L\}$. Now, let $R = \{R_1, R_2, ..., R_n\}$, $C = \{C_1, C_2, ..., C_n\}$, and $L = \{L_1, L_2, ..., L_n\}$. An edge $\{R_i, C_j, L_k\}$ in $\mathcal{G}$, where $i, j, k \in [n]$, is interpreted as being analogous to the entry in row i, column j, and layer k in the latin cube of order n that $\mathcal{G}$ represents. An example of a 1-factorization of $K^3_{2,2,2}$ is shown in figure 12. The latin cube of order 2 that it corresponds to is shown in figure 13.

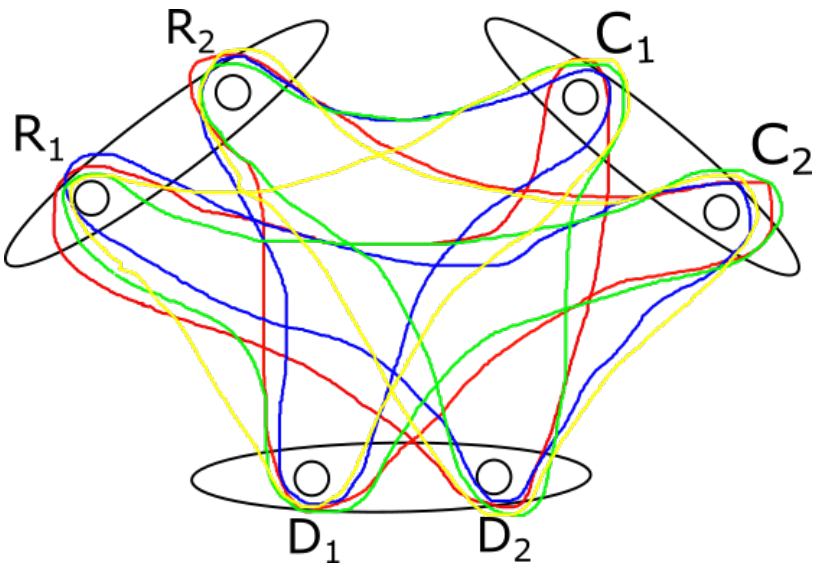

FIGURE 12. A 1-factorization of $K^3_{2,2,2}$ corresponding to the completed latin cube of order 2 shown below in figure 13.

depth 1

A	B
C	D

depth 2

D	C
B	A

FIGURE 13. A completed latin cube of order 2 that corresponds to the 1-factorization of $K^3_{2,2,2}$ shown above in figure 12

4. Ryser's Necessary Condition

Let K be a completed latin square of order n. Suppose K has been partitioned into three mutually disjoint latin rectangles A, B and C. C is of dimension $r \times s$, A is of dimension $r \times (n-s)$ and B is of dimension $(n-r) \times n$. Such a structure is depicted schematically in figure 14.

Let $A(i)$, $B(i)$ and $C(i)$ denote the number of occurrences of symbol i ($i \in [n]$) in A, B and C, respectively. In order for K to be Latin, we need the following three conditions to hold:

(1) $n = A(i) + B(i) + C(i)$
(2) $A(i) \leqslant n - s$
(3) $B(i) = n - r$

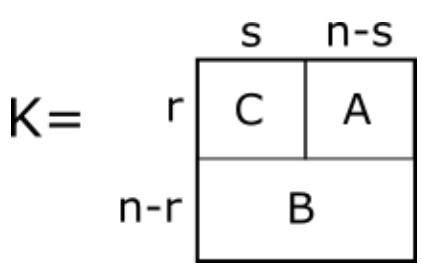

FIGURE 14.

Condition (1) follows unambiguously from the definition of a latin square. The number of occurrences of each symbol in a latin squre of order n is exactly n. Therefore, if we add up the occurrences of a given symbol across all mutually disjoint subsections of a latin square of order n, we must get exactly n.

Condition (2) is the result of the fact that A composes a region of K which is of dimension $r \times (n-s)$. The number of occurrences of a given symbol must not exceed $n-s$. The reason that $A(i)$ is not exactly $n-s$ is because a symbol which is in A may also occur in C while keeping K latin. Additionally, we may say that $A(i) \leqslant r$, but this version is less applicable to a necessary condition on region C of given dimension $r \times s$.

Condition (3) results from the fact that B is of dimension $(n-r) \times n$. In order for K to be latin, the number of occurrences of a given symbol i in B must be exactly $n-r$. The

number of occurrences is not the number of columns n in B, since $n-r \leqslant n$ for all $n, r \in \mathbb{Z}^+$. So $n-r$ is the stricter upper bound.

Putting conditions (1), (2), and (3) together, we have that:

$$n \leqslant C(i) + (n-s) + (n-r)$$
$$C(i) \geqslant r + s - n$$

We may interpret this last inequality as a necessary condition for a latin rectangle of dimension $r \times s$ (corresponding to region C of K) to be extendable to a latin square of order n. The next section will show, using graph theory, that this condition is also sufficient for the extension from $r \times s$ to $n \times n$ to be successful.

5. Ryser's Sufficient Condition

Suppose R is a latin rectangle of dimension $r \times c$ on n symbols where $r, c, n \in \mathbb{Z}^+$ and $r, c < n$. Let $N_R(i)$ denote the number of occurrences of symbol i in R where $i \in [n]$. We look to embed R in a latin square of order n. We know from the previous section that there is no possible way of doing this without requiring that $N_R(i) \geqslant r + c - n$ for $i \in [n]$, so we assume that this condition is met.

Form the bipartite graph $G := G_\rho(R)$ where $V(G) = \{\rho, s\}$. Let $\rho = \{\rho_1, ..., \rho_r\}$ and $s = \{s_1, ..., s_n\}$. An edge in G between ρ_j & s_i is present if and only if symbol i is missing from row j in R, where $j \in [r]$ and $i \in [n]$. Row j of R contains exactly c symbols, therefore $deg_G(\rho_j) = n - c$ for $j \in [r]$. In R, each symbol i has occurred $N_R(i)$ times. Thus, $deg_G(s_i) = r - N_R(i)$ for $i \in [n]$ (this technique of proof is taken from reference [6]). In conjunction with Ryser's necessary condition, this statement becomes:

$$\begin{aligned} deg_G(i) &= r - N_R(i) \\ &\leqslant r - (r + c - n) = n - c \end{aligned}$$

An equivalent way of stating this last inequality is that $\Delta(G) = n - c$. Given this, we borrow a theorem from König (1918) that any bipartite graph F with $\Delta(F) = w$ has a proper edge coloring with w colors[1]. A proper coloring is one where each edge that is incident to a given vertex has a distinct color. This $(n-c)$-proper coloring of G corresponds to an extension of the initial latin rectangle R of dimension $r \times c$ to another latin rectangle of dimension $r \times n$, call it R^*, by adding $n-c$ columns and filling the entries while keeping the structure latin.

This proper $(n-c)$-edge coloring of the bipartite graph representing the rows of R guarantees that a symbol in R appears exactly once in a given row. Once this coloring is done for the rows, it can be done for the columns in the same way, but with an $(n-r)$-coloring. Since no extra restrictions have been imposed, this shows that Ryser's necessary condition, as stated in the the previous section, is also sufficient. Two examples using this method are demonstrated in the next section. The toughest part is finding the correct coloring of the bipartite graph, if it is substantially populated with edges. This can be best seen by observing figure 22.

6. Two examples demonstrating an extension of an $r \times c$ latin rectangle to an $n \times n$ latin cube using graph theory

1	2	3	4	5
5	6	1	3	2
6	1	4	2	3
4	3	2	5	6

Figure 15. Latin rectangle R

Let R be the latin rectangle shown above in figure 15. The right-hand side of Ryser's condition for R reads $4 + 5 - n = 9 - n$. Six distinct symbols are present in R, and the number that a given symbol occurs in R is at least 3. Thus, $N_R(i) = 3 \geqslant 9 - n \rightarrow n \geqslant 6$ implies that the smallest latin square we can extend R to successfully is one of order 6.

A bipartite graph representation of the extension from R to latin rectangle R_6^* of dimension 4×6 is shown below in figure 16, where ρ_i represents the ith row in R. Recall that an edge is present between ρ_i and j iff symbol j is missing from row i. The extension of R to a 4×6 latin rectangle (labeled R_6^*) which corresponds to the coloring of the bipartite graph in figure 16 is shown in figure 17.

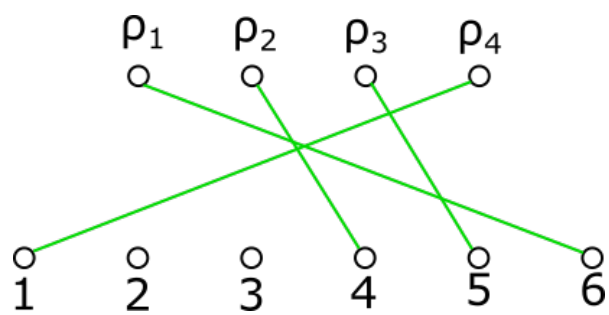

Figure 16. Bipartite graph corresponding to row extension in figure 17

1	2	3	4	5	6
5	6	1	3	2	4
6	1	4	2	3	5
4	3	2	5	6	1

Figure 17. Latin rectangle R_6^*; an extension from R to a 4×6 latin rectangle

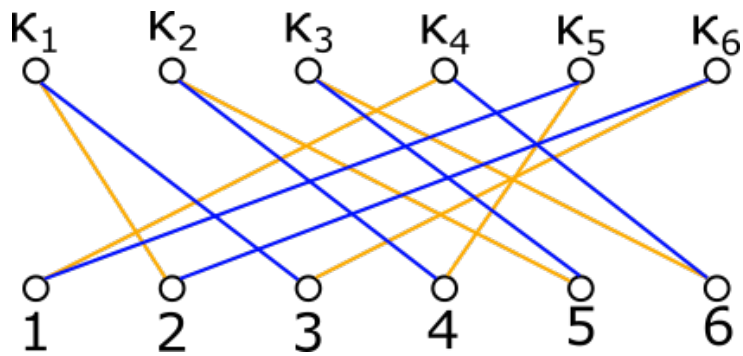

FIGURE 18. Bipartite graph corresponding to column extension in figure 19

1	2	3	4	5	6
5	6	1	3	2	4
6	1	4	2	3	5
4	3	2	5	6	1
2	5	6	1	4	3
3	4	5	6	1	2

FIGURE 19. Final result of extension from R to a latin square of order 6; guided by the graph in figure 18

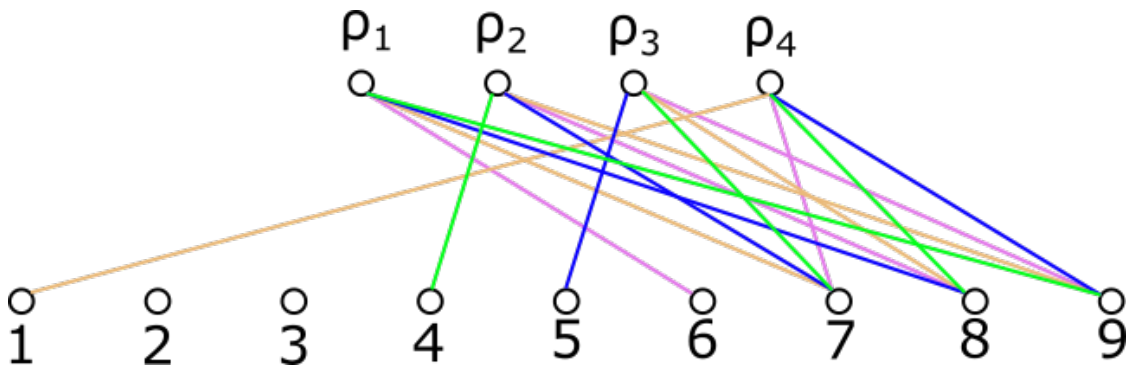

FIGURE 20. Bipartite graph corresponding to row extension in figure 21

Next, we extend R_6^* downward by coloring a bipartite graph that represents the symbols missing from the ith column of R^* (κ_i in the graph). Such a graph is shown in figure 18. The extension of R^* to a latin square of order 6 that corresponds to the colored bipartite graph of figure 18 is shown in figure 19.

This process is also shown for an extension of R to a latin square of order 9 by figures 20 thru 23.

1	2	3	4	5	6	7	8	9
5	6	1	3	2	8	9	7	4
6	1	4	2	3	9	8	5	7
4	3	2	5	6	7	1	9	8

FIGURE 21. Latin rectangle R_9^*; an extension from R to a 4×9 latin rectangle

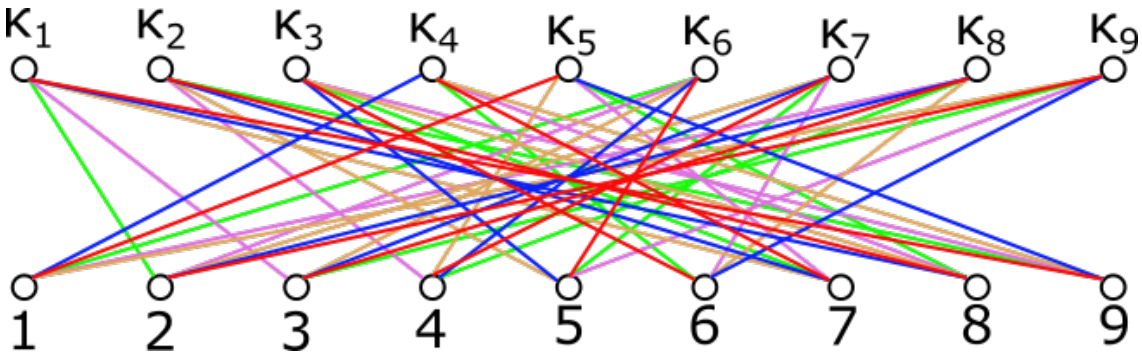

FIGURE 22. Bipartite graph corresponding to column extension in figure 23

1	2	3	4	5	6	7	8	9
5	6	1	3	2	8	9	7	4
6	1	4	2	3	9	8	5	7
4	3	2	5	6	7	1	9	8
2	9	7	6	8	1	5	4	3
3	4	9	8	7	2	6	1	5
7	5	8	9	4	3	2	6	1
8	7	5	1	9	4	3	2	6
9	8	6	7	1	5	4	3	2

FIGURE 23. Final result of extension from R to a latin square of order 9

7. A Ryser-like necessary condition for latin cubes

Let L be the completed latin cube of order n ($n \in \mathbb{Z}^+$) depicted schematically below in figure 24. L has been partitioned into sub-sections: C, A, and B. The subsection of L in the top left of the layers in figure 24 is of dimension $r \times s \times n$. The subsection of L in the top right of the layers in figure 24 is of dimension $r \times (n-s) \times n$. The bottom subsection of L, denoted by B, is of dimension $(n-r) \times n \times n$.

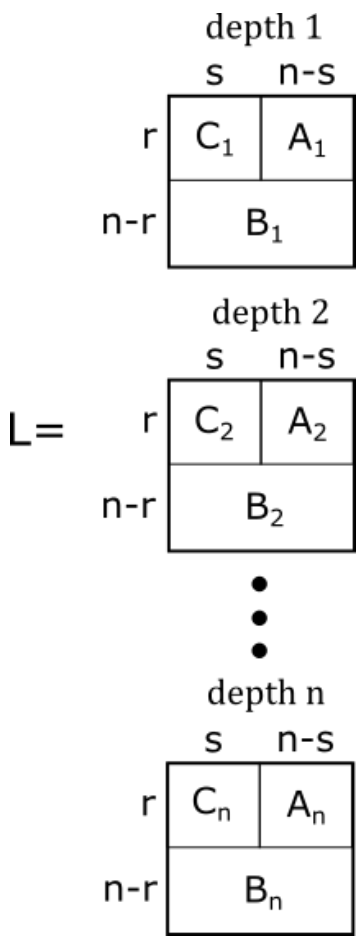

Figure 24.

Let $i \in [n^2]$ and $j \in [n]$. Let $C_j(i)$, $A_j(i)$, and $B_j(i)$ denote the number of occurrences in L of symbol i in depth j of subsections C, A, and B, respectively. In order for L to be latin, the following three conditions must hold for all $i \in [n^2]$:

(1) $\sum_{j=1}^{n} A_j(i) \leqslant n - s$

(2) $\sum_{j=1}^{n} B_j(i) = n - r$

(3) $n = \sum_{j=1}^{n} C_j(i) + \sum_{j=1}^{n} A_j(i) + \sum_{j=1}^{n} B_j(i)$

The reasoning for each of these conditions is analogous to the explanation given in the case of latin squares. Putting conditions (1), (2), and (3) from above together, we have that:

$$n \leqslant \sum_{j=1}^{n} C_j(i) + (n-s) + (n-r)$$

$$\sum_{j=1}^{n} C_j(i) \geqslant r + s - n$$

This last equation may be interpreted as a necessary condition for a latin rectangular prism of dimension $r \times s \times n$ to be embbedable within a latin cube of order n. We will look next to sufficient conditions for a particular case of the latin rectangular prism.

8. The case of an $n \times n \times (n-1)$ latin rectangular prism embedded in an $n \times n \times n$ completed latin cube

When we say that a latin rectangular prism J is of dimension $a \times b \times c$, we mean that J has a rows, b columns, and c depths where $a, b, c \in \mathbb{Z}^+$. The simplest case of extending a partial latin cube to a completed latin cube is when there's only one more slice to fill. An example of extending a $3 \times 3 \times 2$ latin rectangular prism to a $3 \times 3 \times 3$ latin cube is shown in figure 25. An example of extending a partial $4 \times 4 \times 3$ prism to a $4 \times 4 \times 4$ cube is shown in figure 26.

It is not always the case that an $n \times n \times (n-1)$ latin rectangular prism can be embedded in a completed latin cube of order n, which may be seen by the examples in figure 27 & 28. The symbols that are missing in depth 1 of the cube in figure 27 are C, F, and H. Yet, all three of these symbols appear in the third column of the cube. Therefore, there is no way to fill in even the first depth without violating latin-ness. For this reason, we conclude that the latin prism of figure 27 cannot be embedded in a latin cube of order 3. Similarly, depth 1 of the cube in figure 28 is missing the symbols D, F, G, and N. However, all four of these symbols appear in column two of the cube. Thus, the latin rectangular prism in figure 28 cannot be embedded in a completed latin cube.

Unembeddable $n \times n \times (n-1)$ prisms may be modified to make them embeddable in an $n \times n \times n$ latin cube. This is demonstrated with the partial latin cube in figure 27. If we swap the positions of B & E in depth 1 and swap the positions of E & C in depth 3 of the partial cube in figure 27, we are able to fill in depth 1 successfully. Upon filling entries after the swaps, we obtain the partial cube in figure 29. However, there is still an issue with this cube when attempting to fill the other layers. This is because depth 2 of the cube in figure 29 is missing the symbols E, G, and I, yet all three of these symbols appear in column 2 of the cube. Thankfully, another modification makes it extendable. Take the first depth of the cube in figure 29 and swap the symbols A & E. Upon doing so, depths 2 & 3 can be filled in a way that makes the entire structure a completed latin cube. The result is shown in figure 30.

The efforts of the previous two paragraphs convince us that the relationship between positioning of symbols and the specific symbols that are needed in order to complete a given slice are crucial when deciding whether a partial cube may be embedded in a completed cube. In particular, if a given slice of a partial latin cube H of order n requires fn more entries to become filled, where $f, n \in \mathbb{Z}^+$, the exact symbols that this slice of H requires cannot all appear in a slice that is orthogonal to the slice under consideration.

depth 1

A	B	C
E	F	D
I	G	H

depth 2

D	E	F
H	I	G
C	A	B

depth 3

G	H	I
B	C	A
F	D	E

FIGURE 25. An extension of a partial $3 \times 3 \times 2$ latin rectangular prism to a $3 \times 3 \times 3$ latin cube. Black symbols denote entries in the original latin prism. Red symbols denote entries from the extension to the cube.

depth 1

A	B	C	D
F	G	H	E
K	L	I	J
P	M	N	O

depth 2

E	F	G	H
J	K	L	I
O	P	M	N
D	A	B	C

depth 3

I	J	K	L
N	O	P	M
C	D	A	B
H	E	F	G

depth 4

M	N	O	P
B	C	D	A
G	H	E	F
L	I	J	K

FIGURE 26. An extension of a partial $4 \times 4 \times 3$ latin rectangular prism to a $4 \times 4 \times 4$ latin cube. Black symbols denote entries in the original latin prism. Red symbols denote entries from the extension to the cube.

depth 1

A	B	E
D	I	G

depth 2

C	D	H
B	F	A

depth 3

I	G	F
H	E	C

FIGURE 27. A $3 \times 3 \times 2$ latin rectangular prism that cannot be extended to a $3 \times 3 \times 3$ latin cube.

depth 1

A	O	H	C
P	K	E	I
M	J	L	B

depth 2

B	D	G	E
C	F	A	H
I	N	O	P

depth 3

J	I	K	L
D	B	M	N
E	G	F	A

depth 4

N	M	P	F
G	L	J	O
H	C	D	K

FIGURE 28. A $4 \times 4 \times 3$ latin rectangular prism that cannot be extended to a $4 \times 4 \times 4$ latin cube.

depth 1

A	E	B
D	I	G
F	H	C

depth 2

C	D	H
B	F	A

depth 3

I	G	F
H	C	E

FIGURE 29. A modification (highlighted in blue) of the partial latin cube in figure 27, to allow for a successful filling of the first layer (depicted in red)

depth 1

E	A	B
D	I	G
F	H	C

depth 2

C	D	H
B	F	A
G	E	I

depth 3

I	G	F
H	C	E
A	B	D

FIGURE 30. A modification of the partial latin cube in figure 29 (highlighted in blue), to allow for a successful filling of layers 2 & 3 (depicted in red)

9. The general problem

With this definition of latin cube, it appears that the following two questions are logically equivalent:

(1) When can a partial latin rectangular prism of dimension $r \times s \times t$ be embedded in a complete latin cube of order n?

(2) When can a partial 1-factorization of $K^3_{r,s,t}$ be extended to a 1-factorization of $K^3_{n,n,n}$?

A visualization of this equivalence is displayed in figure 31 below. While we are unaware of any theorems regarding 1-factorizations of 3-uniform tripartite graphs in particular, it may be possible to utilize results regarding regular-colorings of complete hypergraphs[7], in the case of a "commutative" latin cube, to shed light on the partial latin cube issue. The reason we say this, is because commutative latin squares of order n are equivalent to 1-factorizations of the complete graph K_n. Commutativity in a latin cube however, may not be as clearly defined as it is in the case of a latin square.

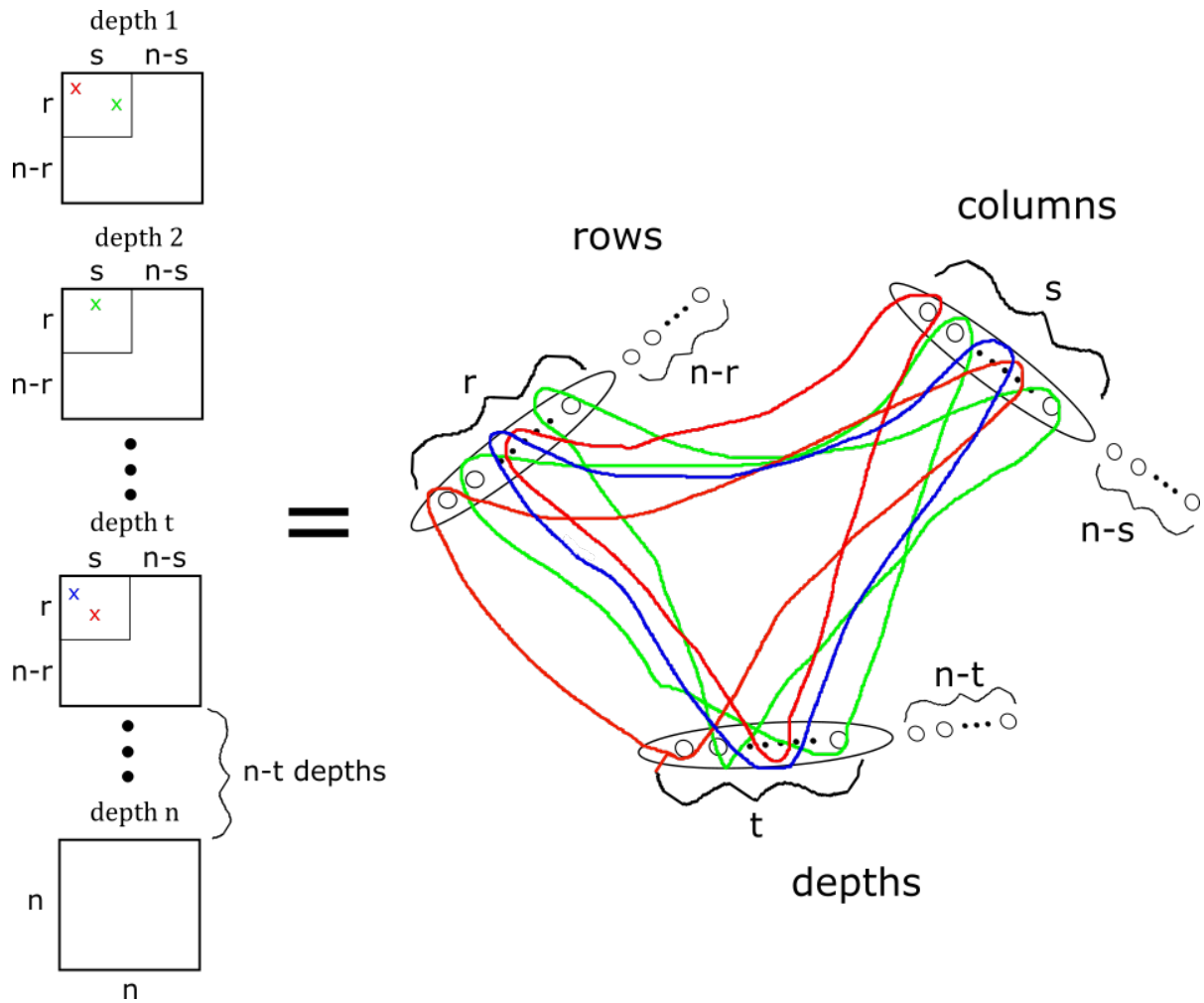

Figure 31. Schematic of the general problem

10. Existing Literature on latin cubes

Various published works examine the conditions on when partial latin cubes can be completed. However, it is worth mentioning that latin cubes have been defined in different ways. A definition of latin cube more restrictive than defined here, is a 3-dimensional array where each slice must be a latin square[5]. This version of latin cube has exactly n symbols, whereas our definition uses n^2 symbols. With this definition, Potapov found that a latin hypercuboid[1] of order 4 can be extended to a latin hypercube of order 4.

Bryant et al. (2012) defined an $n \times n \times k$ latin cuboid as a 3-dimensional array containing n distinct symbols such that each symbol occurs once in each row, column, and depth. They found that[4]:

(1) for $m \geqslant 2$, no $2m$ x $2m$ x m awkward[2] latin cuboid is completable.
(2) for $m \geqslant 8$, there exists a noncompletable $2m$ x $2m$ x m latin cuboid which is extendable to a $2m$ x $2m$ x $(m+1)$ latin cuboid.
(3) for all even $m \geqslant 2$, there exists a $(2m-1)$ x $(2m-1)$ x $(m-1)$ nonextendible latin cuboid.

Additionally, Denley & Ohman (2014) define a $k \times l \times m$ latin box as a $k \times l \times m$ partial latin cube where all the cells are filled. They argue that[3]:

(1) given a k x l x m latin box, if $m + 2l + 2k - 4 \leqslant n$ then it can be completed to a $k \times l \times n$ latin box.
(2) given a $2 \times 2 \times m$ partial latin cube, if $m \leqslant n-4$ then it can be extended to a $2 \times 2 \times n$ partial latin cube.
(3) for $n \geqslant 14$, any $2 \times 2 \times n$ latin box can be completed to a $2 \times 2 \times n$ latin box

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[1]The table of values of an n-ary quasigroup is called a latin hypercube (the n-dimensional generalization of latin square) and the table of values of a partial n-ary quasigroup is called a latin hypercuboid. A function $f : \sum^n \to \sum$ is called an n-ary quasi group if $f(\overline{x}) \neq f(\overline{y})$ for all $\overline{x}, \overline{y} \in \sum^n$ such that $d(\overline{x}, \overline{y}) = 1$ where $d(x, y)$ is the Hamming distance (number of positions in which $\overline{x}$ and $\overline{y}$ differ)

[2]Special case of latin cubes